

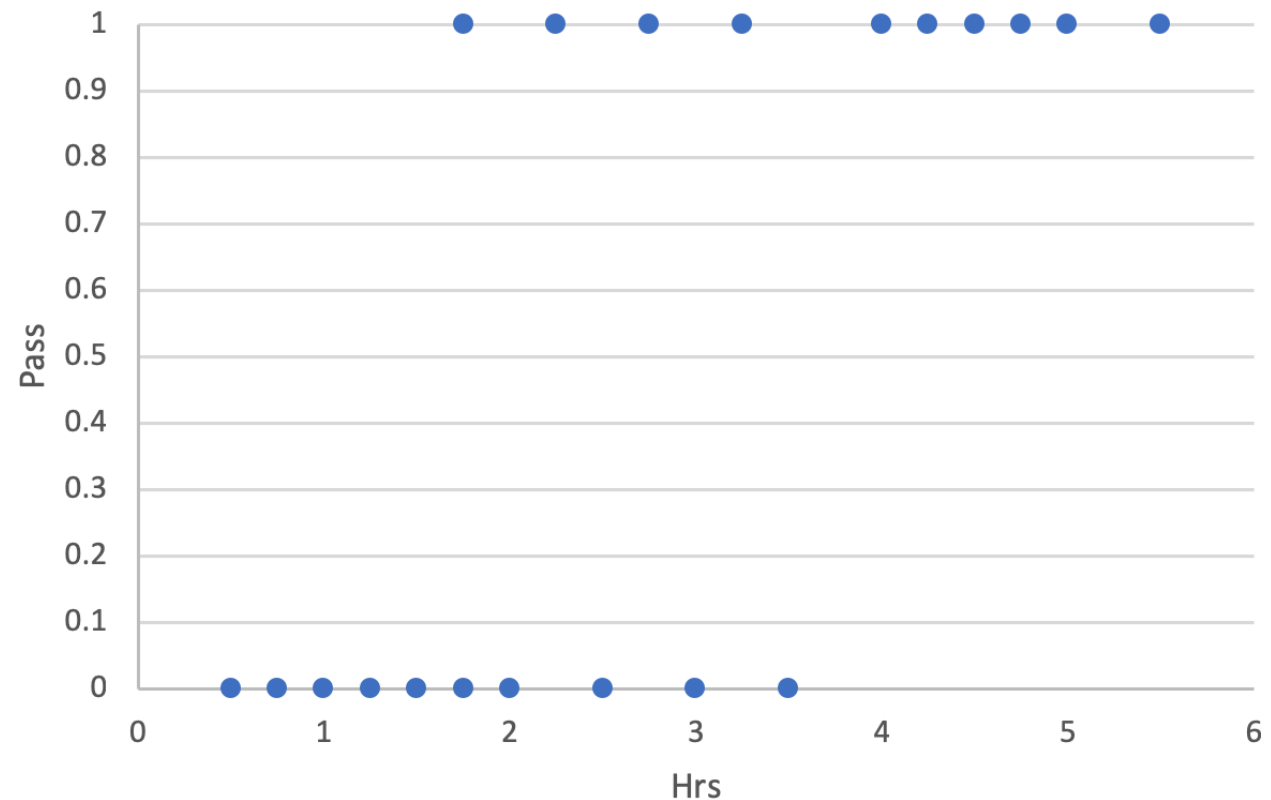
Machine Learning

Logistic Regression

Linear Regression for Classification?

- How likely is a student to pass if he/she studies for 5 hrs?
- Using data from 20 students
- Classification problem! Can use linear regression?

Hrs	Pass?
0.5	0
0.75	0
1	0
1.25	0
1.5	0
1.75	0
1.75	1
2	0
2.25	1
2.5	0
2.75	1
3	0
3.25	1
3.5	0
4	1
4.25	1
4.5	1
4.75	1
5	1
5.5	1

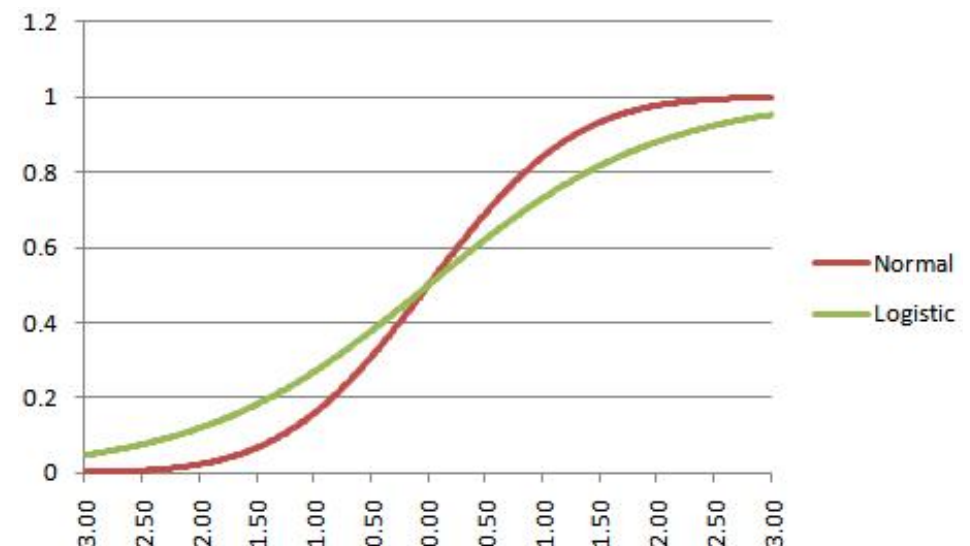


Instead can we fit a curve?

- Regression fits $y = a + bx$
- Instead why not fit?

$$y = f(a + bx)$$

- Common choices for $f()$
 - Logistic Regression: $y = \frac{1}{1 + e^{-(a+bx)}}$
 - Probit Regression: $y = \Phi(x)$



The Logit function

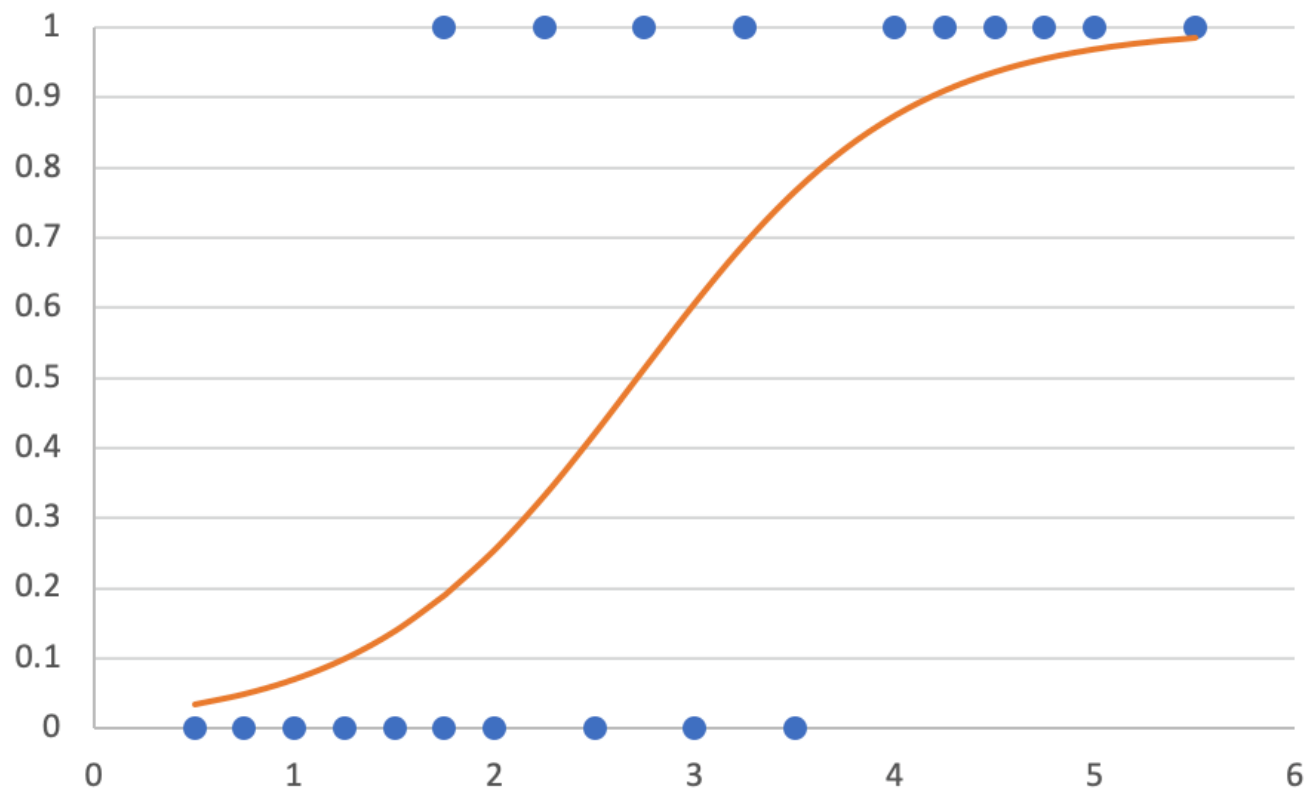
- Logit function: $y = \frac{1}{1 + e^{-(a+bx)}}$
- Equivalent to thinking of $\log \left(\frac{y}{1-y} \right) = a + bx$

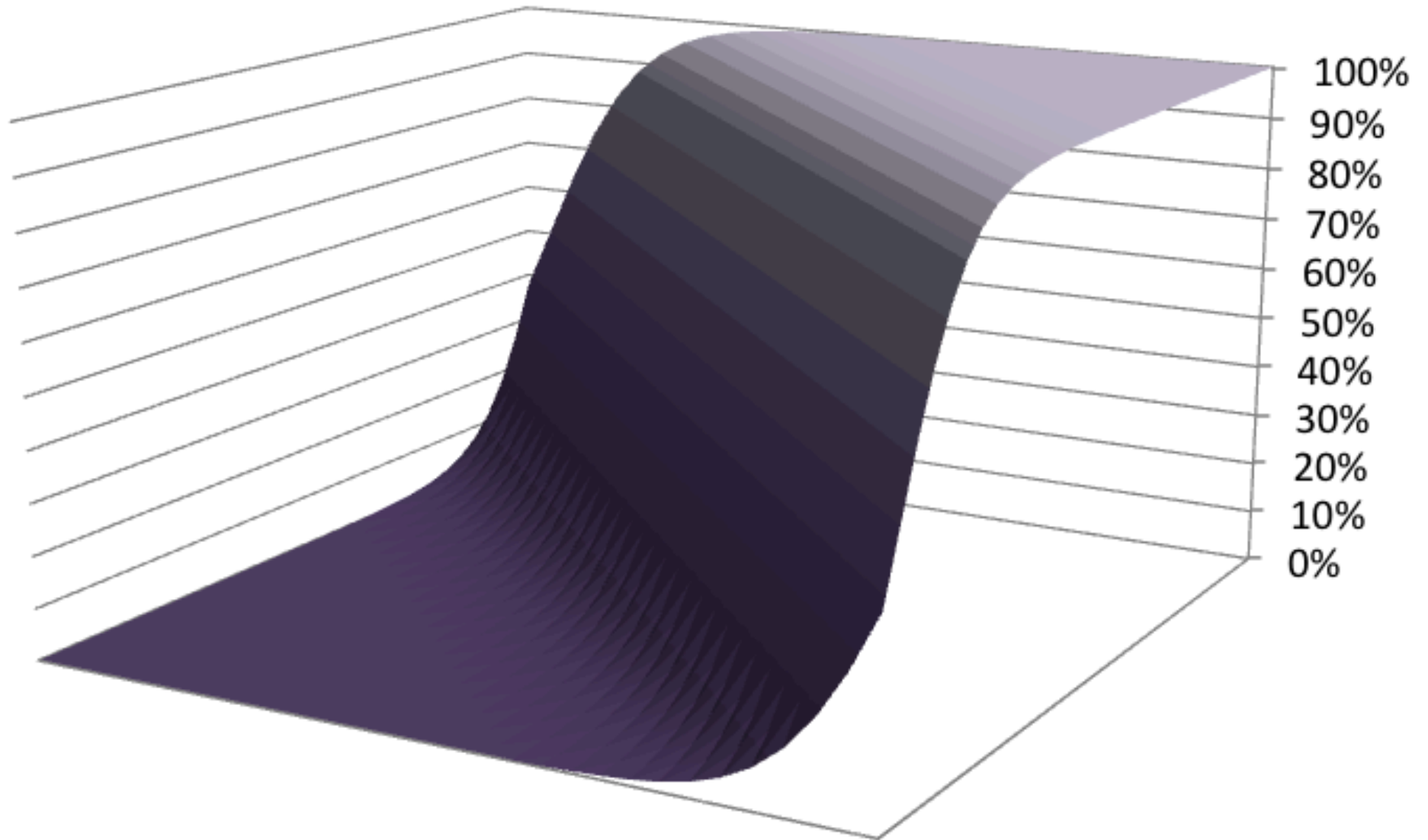
Finding the best fit logic curve?

- Linear regression minimized sum of squared residuals. This unfortunately will not work in logistic regression!
- Instead we choose to minimize the “Log Loss” or “Cross-Entropy”

$$-y \log(\hat{y}) - (1 - y) \log(1 - \hat{y})$$

- How likely is a student to pass if he/she studies for 5 hrs?





Logistic Reg - Pros and Cons

- Advantages
 - A classification model that does give probabilities
 - Easily extended to multiple classes (multinomial regression)
 - Quick to train and very fast at classifying unknown records
- Disadvantages
 - Constructs linear boundaries
 - Assumes that variables are independent (eg. does not include interaction terms)
 - Interpretation of coefficients is difficult